INCH-POUND

MIL-PRF-64266/10B
w/AMENDMENT 1
17 June 2019
SUPERSEDING
MIL-PRF-64266/10B
23 June 2014

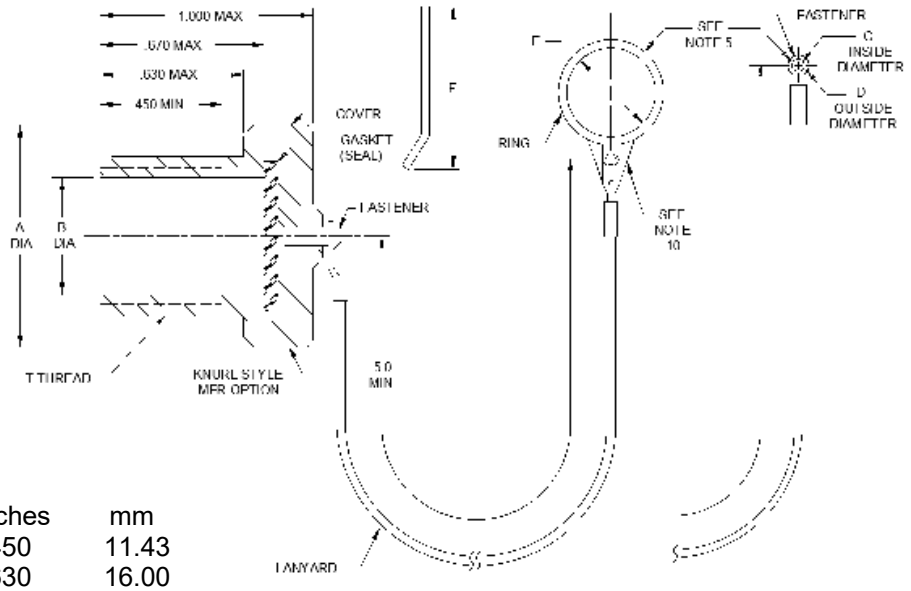
PERFORMANCE SPECIFICATION SHEET

CONNECTORS, FIBER OPTIC, CIRCULAR, PLUG STYLE, MULTIPLE REMOVABLE TERMINI, DUST COVER, SCREW THREADS, ENVIRONMENT RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring fiber optic connectors described herein shall consist of this specification sheet and MIL-PRF-64266.

Dust covers for MIL-PRF-64266 receptacles shall be in accordance with MIL-PRF-64266/11.



Inches	mm
.450	11.43
.630	16.00
.670	17.02
1.000	25.40

FIGURE 1. Plug dust cover.

AMSC N/A

FSC 6060

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Shell size	T thread Class 2A	A dia max		B dia		C dia inside 6/ 7/		D dia outside 6/ 7/		E dia inside		F Max	
		Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
11	.750-.1P-.2L-D.S.	1.028	26.11	.581 .569	14.75 14.45					.608 .578	17.3 14.7	1.5	38.1
13	.875-.1P-.2L-D.S.	1.141	28.98	.705 .693	17.90 17.60	.125 .110	3.17 2.79	.210 .170	5.44 4.32	.773 .703	19.6 17.9	1.7	43.2
15	1.062-.1P-.2L-D.S.	1.263	32.08	.877 .865	22.27 21.97					.905 .875	24.1 22.2	1.8	45.7
23	1.500-.1P-.2L-D.S.	1.705	43.26	1.299 1.287	32.99 32.69	.152 .173	3.87 3.48	.280 .240	7.11 6.10	1.358 1.328	34.5 33.7	2.3	58.4

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions apply to plated/finished part.
4. The connector plug interface dimensions shall be in accordance with figure A-1 of MIL-PRF-64266.
5. Optional configurations (see PIN). Attachment to a backshell can be either by a ring or a fastener; however, one must be specified.
6. Fastener hole diameter is sized for captive screws fastened using a 3/32 inch (2.38 mm) hex wrench for shell sizes 11, 13, and 15.
7. Fastener hole diameter is sized for captive screws fastened using a 7/64 inch (3.57 mm) hex wrench for shell size 23.
8. Ring shall be solid, not split. Ring may contain a sufficient angle at the lanyard transition enabling the ring to sit flush once installed to prevent scratching the paint on the connectors and other mounting surfaces.
9. Usage. The lanyard attachment method of lanyard with (solid backshell) ring is used when the ring is placed on the connector rear accessory threads and tightened against the backshell (or other connector accessory) coupling nut. The lanyard attachment method of lanyard with fastener is used when the fastener is placed on a separate screw in the backshell to retain the dust cover.
10. Ring and fastener shape. Attachment device shape is optional.

FIGURE 1. Plug dust cover - Continued.

REQUIREMENTS:

Dimensions and configurations: See figure 1 herein and MIL-PRF-64266, figure A-1.

Weight: Weight shall not exceed the specified value in table I for the applicable shell size. Weight includes lanyard and ring.

TABLE I. Dust cover weights.

Shell size	Composite		Aluminum		Stainless steel	
	ounces	grams	ounces	grams	ounces	grams
11	0.28	8.0	0.49	14.0	1.27	36.0
13	0.35	10.0	0.56	16.0	1.59	45.0
15	0.42	12.0	0.71	20.0	1.94	55.0
23	0.63	18.0	1.06	30.0	3.17	90.0

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Qualification: Qualification shall consist of performing testing specified as listed in table II. The tests in this initial qualification table are applicable when performed only for the dust covers. When testing the connector plug and connector receptacle, the dust cover shall be included as part of the initial qualification of these components and not tested separately. In either case, the dust cover shall meet the requirements for the same inspections as listed in table II.

TABLE II. Qualification inspections.

Test performed 7/, 11/	Initial qualification	For change in				
		Seal material	Sealing integrity	Plating	Shell size	Lanyard
	1/	2/	3/	4/	5/	6/
GROUP 1 (6 mated pairs)						
Interoperability (not applicable)						
Visual & Mechanical						
Size	X			X	X	
Weight	X			X	X	
Identification marking	X			X	X	
Screw threads	X			X		
Workmanship	X	X	X	X	X	X
Functional						
Dust cover engagement and disengagement torque	X	X	X	X	X	
Dust cover mating durability	X	X	X	X		
Dust cover engagement and disengagement torque	X	X	X	X		
Optical						
None						
GROUP 2 (2 mated pairs)						
Connector rear threads accessory attachment						
Cable pull out force (retention)						
External bending moment						
Cable seal flexing						
Twist						
Impact	X				X	X
Crush	X		X		X	
Vibration: Swept sine						
Vibration: Random						
Shock: MIL-DTL-901 12/	X					X
Water pressure 14/	X	X	X	X	X	
Modified SO_2/salt spray	X	X	X	X		X
Dust cover engagement and disengagement force	X	X	X	X		X

See notes at end of table.

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TABLE II. Qualification inspections – Continued.

Test performed 7/, 11/	Initial qualification	For change in				
		Seal material	Sealing integrity	Plating	Shell size	Lanyard
	1/	2/	3/	4/	5/	6/
GROUP 3 (2 mated pairs)						
Thermal shock 15/	X	X		X		X
Temperature/humidity cycling						
Temperature cycling						
(Temperature life) 15/	X	X		X		X
Freezing water						
Sand & dust	X	X	X	X		
Identification marking	X	X		X		
Dust cover engagement and disengagement torque	X	X	X	X		
Group 4 (2 mated pair + parts) 7/						
Electromagnetic effects (2 mated pair) 9/ 13/	X			X		
Fluid immersion (2 mated pair) 15/	X	X		X		X
Shell to shell conductivity (initial)	X			X		
Salt spray (2 mated pair) 8/	X	X		X		
Shell to shell conductivity (post salt spray)	X			X		
Dust cover engagement and disengagement torque	X	X		X		
Flammability (1 mated pair)						
Fungus resistance (parts) 10/	X	X				X
Ozone exposure (parts) 10/	X	X				X

1/ Test sample configuration. Unless otherwise specified, the connector plug dust cover shall be mated to the connector receptacle dust cover for each of the tests/inspections. When testing the connector plug and connector receptacle as part of the initial qualification, one end of each dust cover lanyard shall be affixed to the connector.

2/ Seal material. Tests in this column may be excluded if other components that use the same seal material are qualified to this defense specification.

3/ Seal integrity. These tests verify the sealing integrity, verify aspects unique to the dust cover configuration, and so shall be performed as part of a dust cover qualification for a change in seal material or dust cover design.

4/ Plating. Tests in this column other than those for group 1 visual and mechanical inspection may be excluded if other components that use the same plating are qualified to this defense specification with one provision: The other components must contain the same knurl or other style gripping surface as dust covers being qualified.

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TABLE II. Qualification inspections – Continued.

5/ Shell size. With the exception of those tests listed in group 1, tests in this column can be excluded for smaller shell sizes once dust covers for larger shell sizes are qualified to this defense specification.
6/ Lanyard attachment. Lanyard attachment shall consist of a coated stainless steel wire rope with ring. This column applies when a material or configuration (such as attachment method to dust cover) is changed.
7/ Group 1 mated pair are to be used for Groups 2 and 3 tests. Group 4 may be performed before Group 1, and, except for electromagnetic effects, with separate samples.
8/ Salt spray. Two options: a. Use same two mated pair from the fluid immersion test. b. Use separate mated pair.
9/ Electromagnetic effects. Two (2) mated pair from the fluid immersion, salt spray, or Group 2/3 samples after that Group's test completion may be used.
10/ Parts only, assembly not required.
11/ Specific test practices for physical, mechanical, environmental and material tests, including clarifications and further details, are found in MIL-STD-1678-3.
12/ Shock test. Mated dust covers shall be affixed to standard shock fixture 4A for bulkhead mounting in the same manner as the dust covers are affixed to the connector. There shall be no damage or loosening of parts and mated dust covers shall not become a missile hazard.
13/ For receptacle dust cover only. Two of the six mated pair from Group 1 shall be used. Test sample configuration shall consist of a receptacle dust cover mated to a connector receptacle.
14/ Water pressure. When test is performed with a dust cover mated to a connector and the dust cover is removed for the post test water penetration inspection, orient the connector so that it is facing away from the center of the earth. This orientation will prevent water that may have penetrated threads prior to seals from entering the connector (resulting in a false positive for water penetration).
15/ Temperature ranges 1 and 2. Dust covers may be qualified for either or both temperature range 1 and temperature range 2. For each temperature range qualified, the thermal shock, temperature life, and fluid immersion tests for that temperature range shall be performed.

Maintenance aging: Not applicable.

Twist: Not applicable.

Mating durability: Not applicable.

Dust cover mating durability: Applicable.

Water pressure: Applicable for mated dust covers.

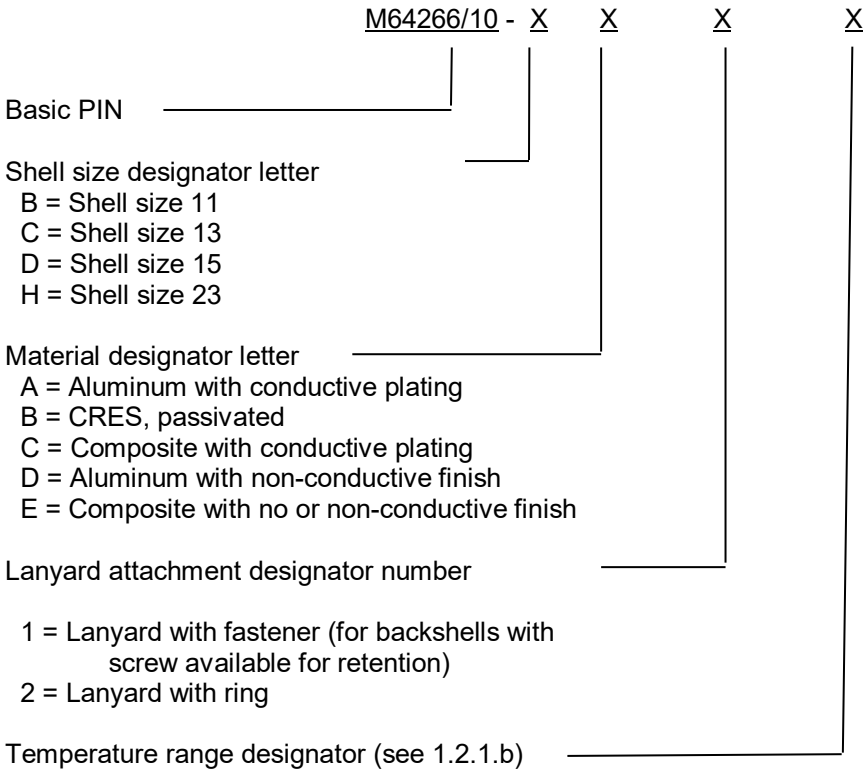
Gasket: Gasket shall be bonded to cover or mechanically retained.

Lanyard: Lanyard shall be free to rotate on fastener affixed to dust cover. A fastener or a ring, as specified in the PIN, shall be affixed to other end of lanyard than the end for the dust cover. Lanyard attachment shall consist of a coated stainless steel wire rope with fastener or with ring. The outside diameter of the ring at a right angle to the crimp sleeve shall meet the values specified for dimension E (see figure 1). The diameter of the coated stainless steel wire rope portion of the lanyard shall not exceed 0.08 inches (2 mm).

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Marking:

Part or Identification Number (PIN): Marked dust cover.



PIN example: M64266/10-BA21

NOTE: Shell size designators in MIL-PRF-64266 conform to those in MIL-DTL-38999 connector (Shell size 11 =B, shell size 13 = C, shell size 15 = D, shell size 23 = H). The shell size designators are not consistent with those in MIL-DTL-28840 or in MIL-PRF-28876 connectors (Shell size 11 =A, shell size 13 = B, shell size 15 = C, shell size 23 = F). Care should be taken when specifying the PIN for those use to specifying the PIN for MIL-DTL-28840 and MIL-PRF-28876 connectors.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-64266, this specification sheet references the following document:

MIL-DTL-28840
MIL-PRF-28876
MIL-DTL-38999
MIL-PRF-64266/11
MIL-STD-1678-3
MIL-DTL-901

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6060-2018-011)

Review activities:
Army - MI
Navy - AS
Air Force - 13, 19, 93
NASA - NA

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